Data Science
Feature Scaling Technique

Presented By : Harsh Dalal

We Make You Trust Data

Agenda

- **Understand the need for Feature Scaling** and its impact on machine learning model performance.
- **Differentiate between Standardization and Normalization** conceptually and mathematically.
- **Apply Min–Max Scaling and Z-Score Standardization** using Python (scikit-learn).
- **Analyze the effect of scaling on data distribution and model behavior.**
- **Implement hands-on feature scaling techniques** on sample datasets and interpret results.

Precap : Foundation Concepts Before Feature Scaling

- **Data Analysis:**
  Understanding features, observations, and data types along with calculating mean, standard deviation, minimum, and maximum to analyze central tendency and dispersion.
- **NumPy & Pandas:**
  Using NumPy for numerical computations and Pandas Data-Frame for structured data manipulation, filtering, and transformation.
- **Matplotlib & Seaborn:**
  Visualizing data distributions to understand spread, patterns, and relationships between variables.

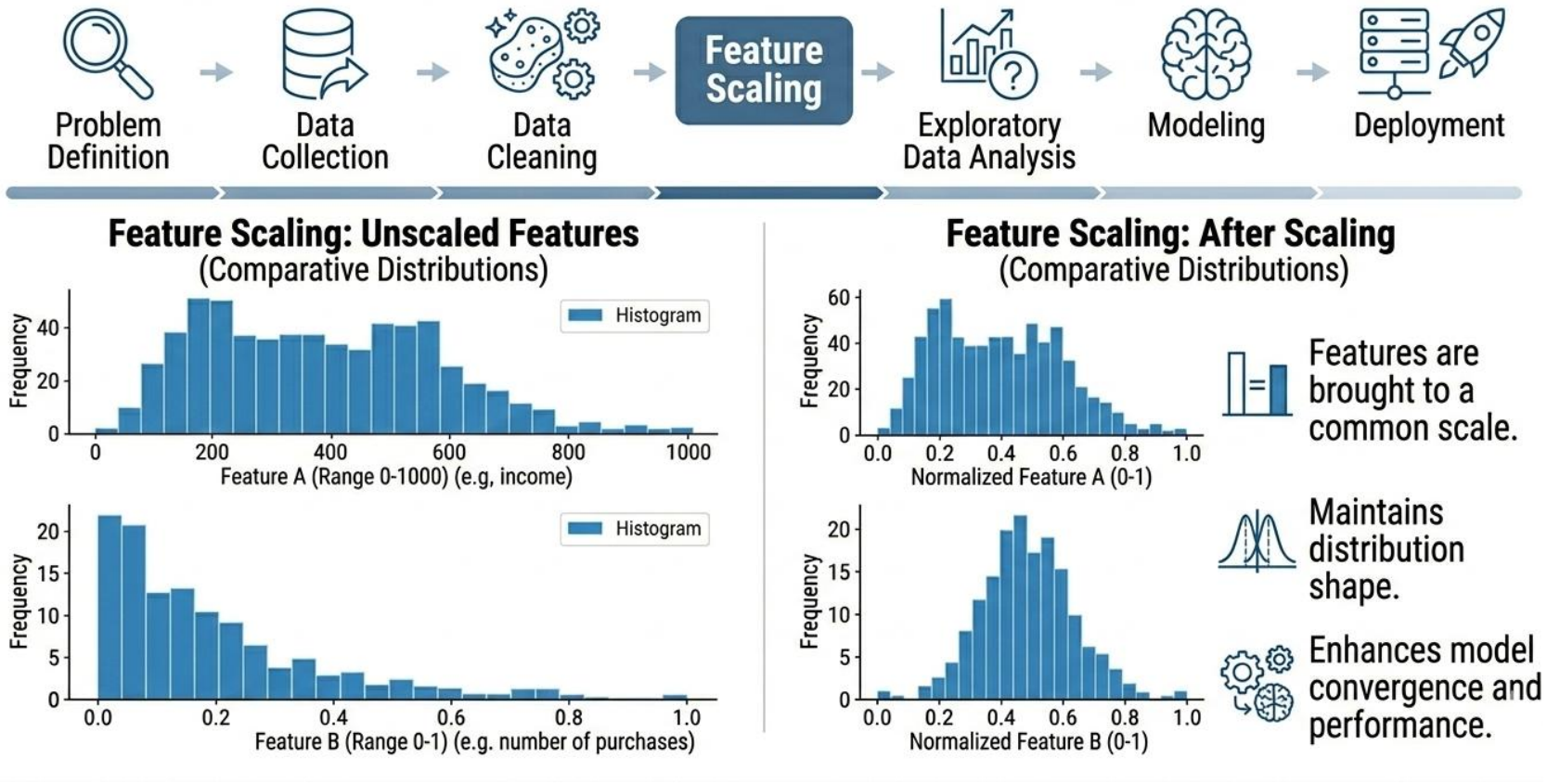
Introduction: Why is Feature Scaling Required?

Feature Scaling ensures that all input variables contribute proportionately to the model by transforming them to a comparable scale.

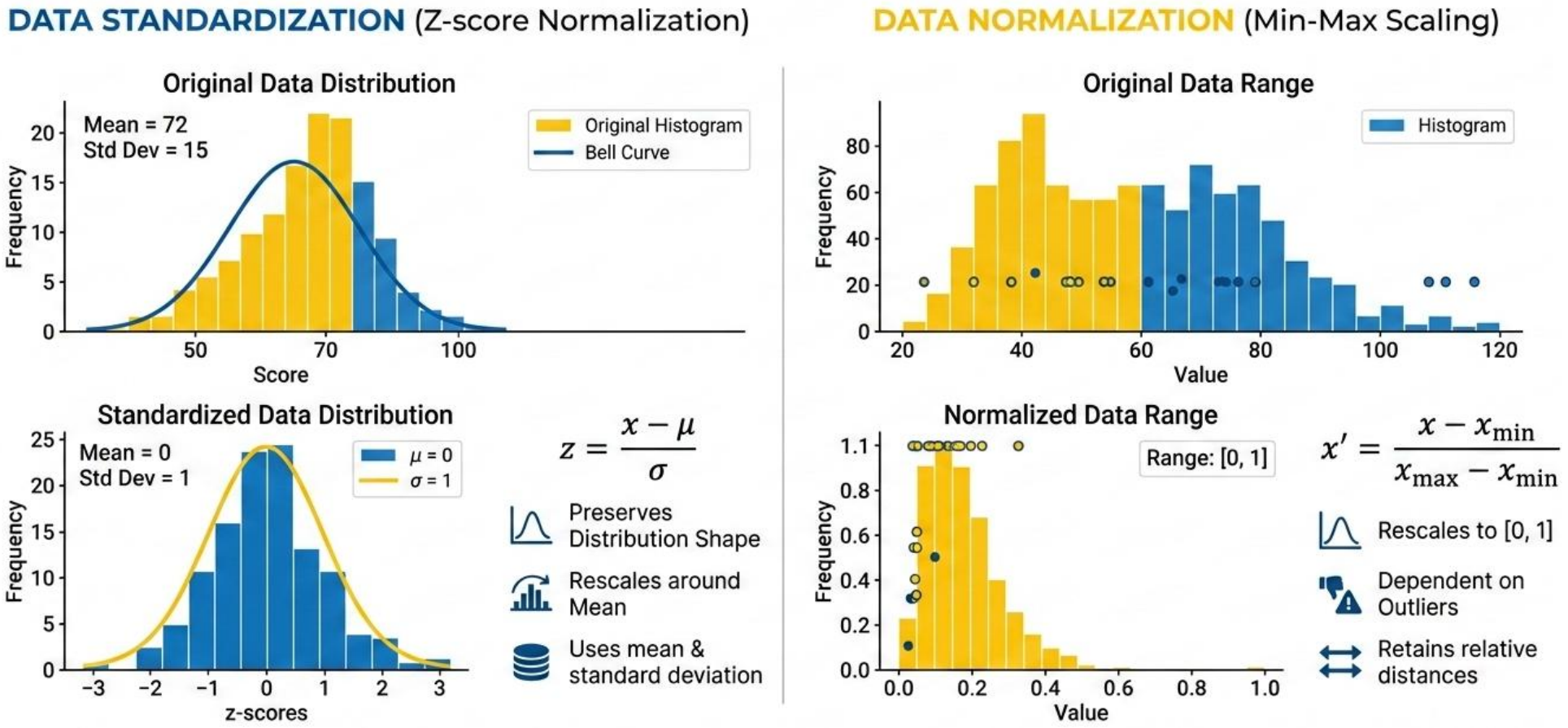
Why It Is Important?

- **Prevents Dominance of Large-Scale Features**
  Variables with larger numeric ranges (e.g., Salary) can overpower smaller ones (e.g., Experience).
- **Improves Distance-Based Algorithms**
  Algorithm like **KNN, K-Means, SVM** rely on distance metrics (Euclidean distance). Uneven scales distort distance calculation.
- **Faster Gradient Descent Convergence**
  In optimization-based models (Linear/Logistic Regression, Neural Networks), scaling ensures smoother and faster convergence.

Before VS After

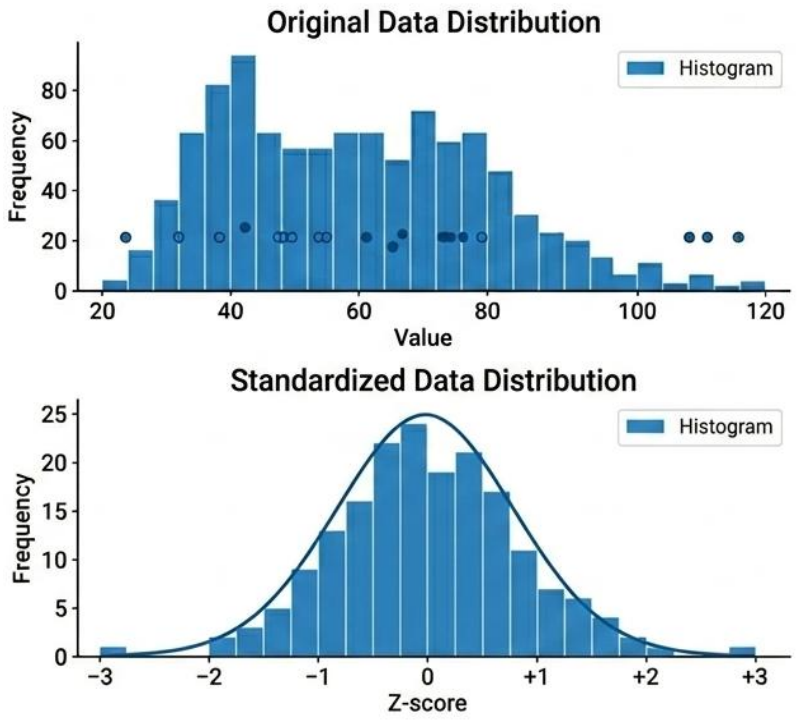


Feature Scaling Techniques



Data Standardization (Min–Max Scaling)

- **Data Standardization** (Z-score Scaling) is a feature scaling technique that transforms data so that it has:
- **Mean (μ) = 0 , Standard Deviation (σ) = 1**
- It centers the data around zero and scales it based on variability.



$$Z = \frac{x - \mu}{\sigma}$$

μ = Mean
σ = Standard Deviation

Transforms data to have a Mean of 0 a Standard Deviation of 1

Less sensitive to Outliers than Min-Max

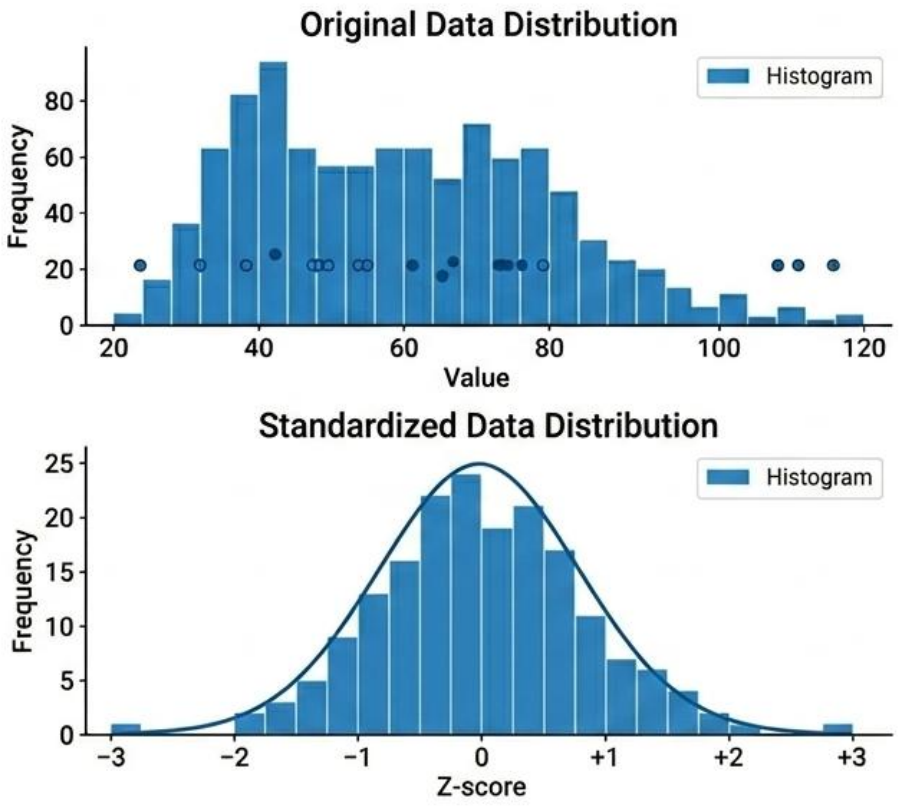
Preserves shape of the original distribution (if normal)

Data Standardization

- **Data Standardization** (Z-score Scaling) is a feature scaling technique that transforms data so that it has:

Mean (μ) = 0 , Standard Deviation (σ) = 1

- It centers the data around zero and scales it based on variability.



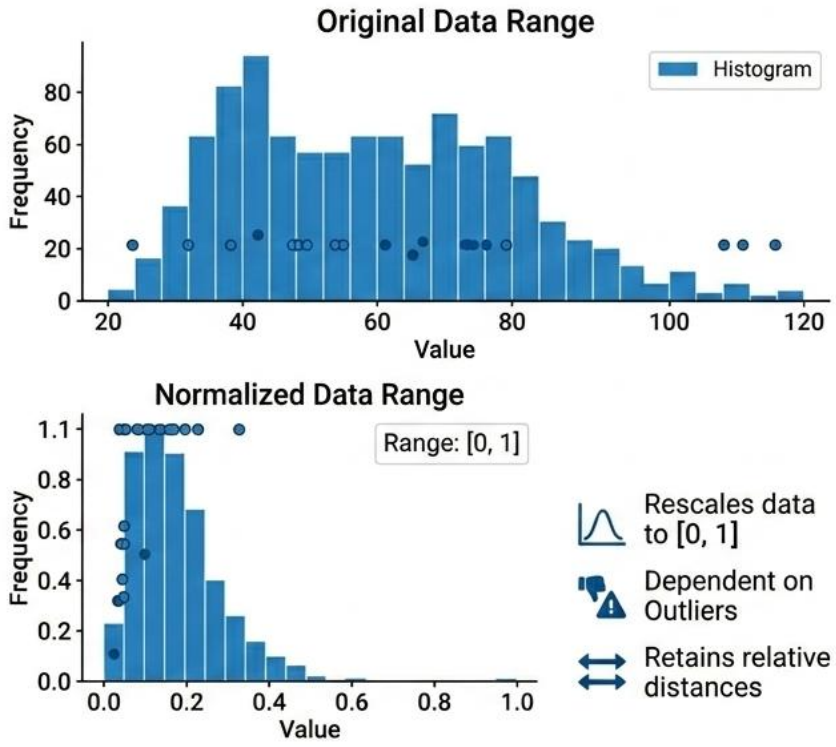
$$Z = \frac{x - \mu}{\sigma}$$

μ = Mean
σ = Standard Deviation

Transforms data to have a Mean of 0 a Standard Deviation of 1

Less sensitive to Outliers than Min-Max

Preserves shape of the original distribution (if normal)



$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Range: [0, 1]

Rescales data to [0, 1]

Dependent on Outliers

Retains relative distances

Data Normalization(Min–Max Scaling)

Data Normalization is a feature scaling technique that transforms numerical values into a fixed range, typically **0 to 1**,without changing the original distribution shape.

UNDERSTANDING LABEL ENCODING & ONE-HOT ENCODING

Raw Categorical Data: ['Red', 'Green', 'Blue', 'Green']

1. LABEL ENCODING

Converts categories to unique integer labels.

Categories: ['Red', 'Green', 'Blue']

Mapping: {'Red': 0, 'Green': 1, 'Blue': 2}

Result: [0, 1, 2, 1]

2. ONE-HOT ENCODING

Unique Categories: ['Red', 'Green', 'Blue']

Columns Created: (3 columns) category.

Matrix:

	Red	Green	Blue
Red:	1	0	0
Green:	0	1	0
Blue:	0	0	1
Green:	0	1	0

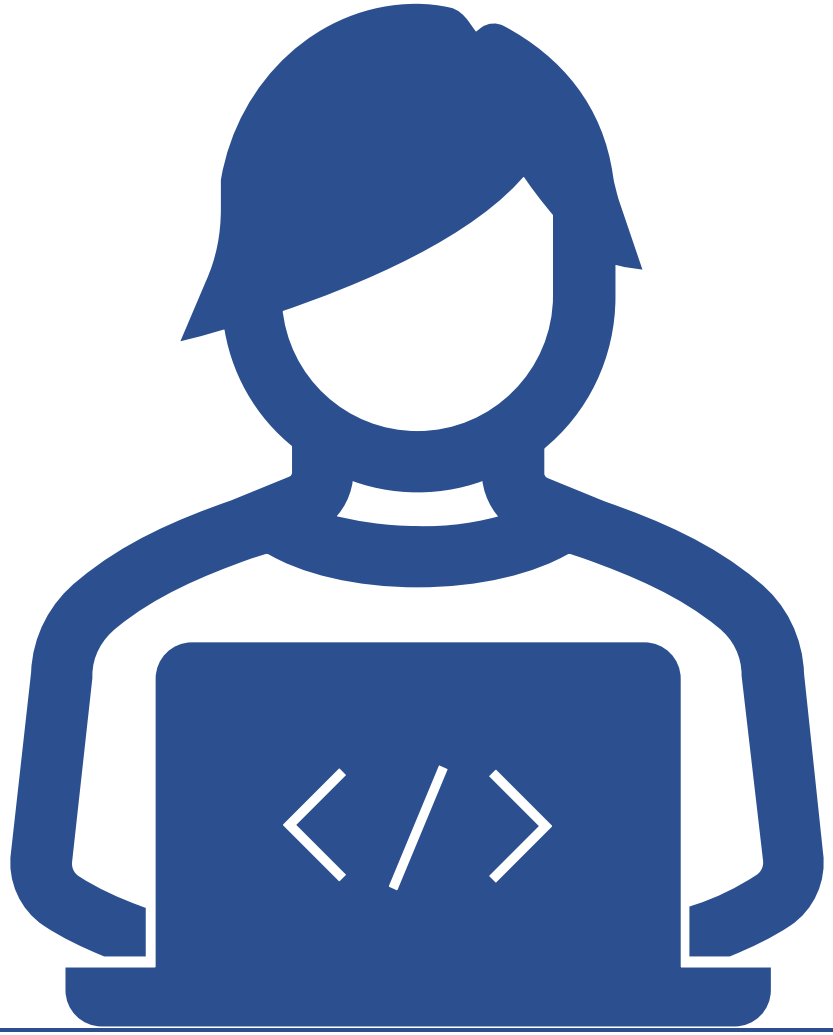
Symmetrical Data

Label Encoding → **Ordinal data:** order is preserved **(0<1<2)**

One-Hot Encoding → **Nominal data:** no intrinsic order **(Red ≠ Green ≠ Blue)**

Correct data representation for different data types.

Hands - on



Summary

- **Feature Scaling** ensures all features are on a similar scale so that machine learning algorithms are not biased toward variables with larger numerical values.
- **Normalization** rescales data to a fixed range (usually **0 to 1**) and is useful when the dataset does not follow a normal distribution.
- **Standardization** transforms data to have **mean = 0 and standard deviation = 1**, which is important for algorithms that assume Gaussian distribution (e.g., Logistic Regression, SVM).
- These techniques improve **model performance, faster convergence, and fair feature contribution** in machine learning models.

Any Query

Thank You

Happy learning

Data Science
Statistical Inference and Probability

Presented By : Harsh Dalal

We Make You Trust Data

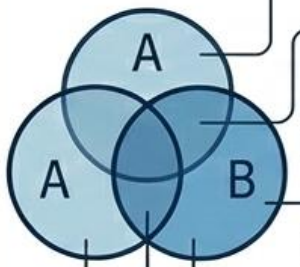
Agenda

- Introduction to **probability fundamentals**, including event relationships and concepts of independent, dependent, and mutually exclusive events.
- Understanding **decision-making under uncertainty** using statistical thinking and probability concepts.
- Learning the **hypothesis testing framework**, including formulation of null and alternative hypotheses with one-tail and two-tail tests.
- Exploring statistical methods such as **Chi-Square test for categorical relationships and ANOVA for comparing multiple groups**.

INTRODUCTION TO PROBABILITY FUNDAMENTALS IN DATA SCIENCE

PROBABILITY RULES & FOUNDATIONS

FOUR KEY PILLARS:



- **1. PROBABILITY RULES:** P(A), P(AUB) — Foundations of Chance, Sample Space, Axioms, Marginal, Joint, Conditional
- **2. RANDOM VARIABLES & DISTRIBUTIONS:** X, PDF, PMF — Quantifying Uncertainty, Discrete (Countable), Continuous (Infinite, Uncountable)
- **3. KEY MEASURES:** E(X), σ^2, σ — Summarizing Complex Data, Expected Value (Mean), Variance, Standard Deviation
- **4. BAYES' THEOREM:** Bayes Rule — Updating Predictions with Evidence
  Prior, Posterior, Updating existing knowledge with new data

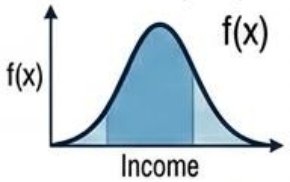
THE MATHEMATICAL LANGUAGE OF UNCERTAINTY

RANDOM VARIABLES, PMF, & PDF

DISCRETE: Probabiiity Mass Function (PMF)

Gives probability for for a specific value

CONTINUOUS: Probability Density Function (PDF)



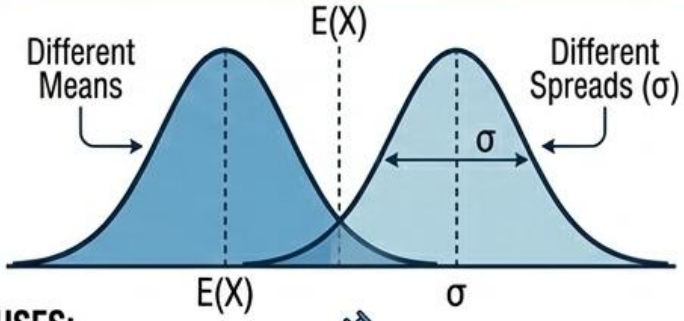
Relative likelihood within a range, not specific point

HOW DATA BEHAVES & CLUSTERS

Example application (Predicting app usage):

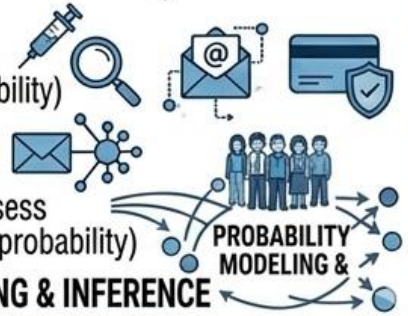
1. Define Random Variable X = # of logins
2. Find probabilities for each value (PMF)
3. Use total PMF sum = 1

MEASURES & PROBABILITY IN PRACTICE

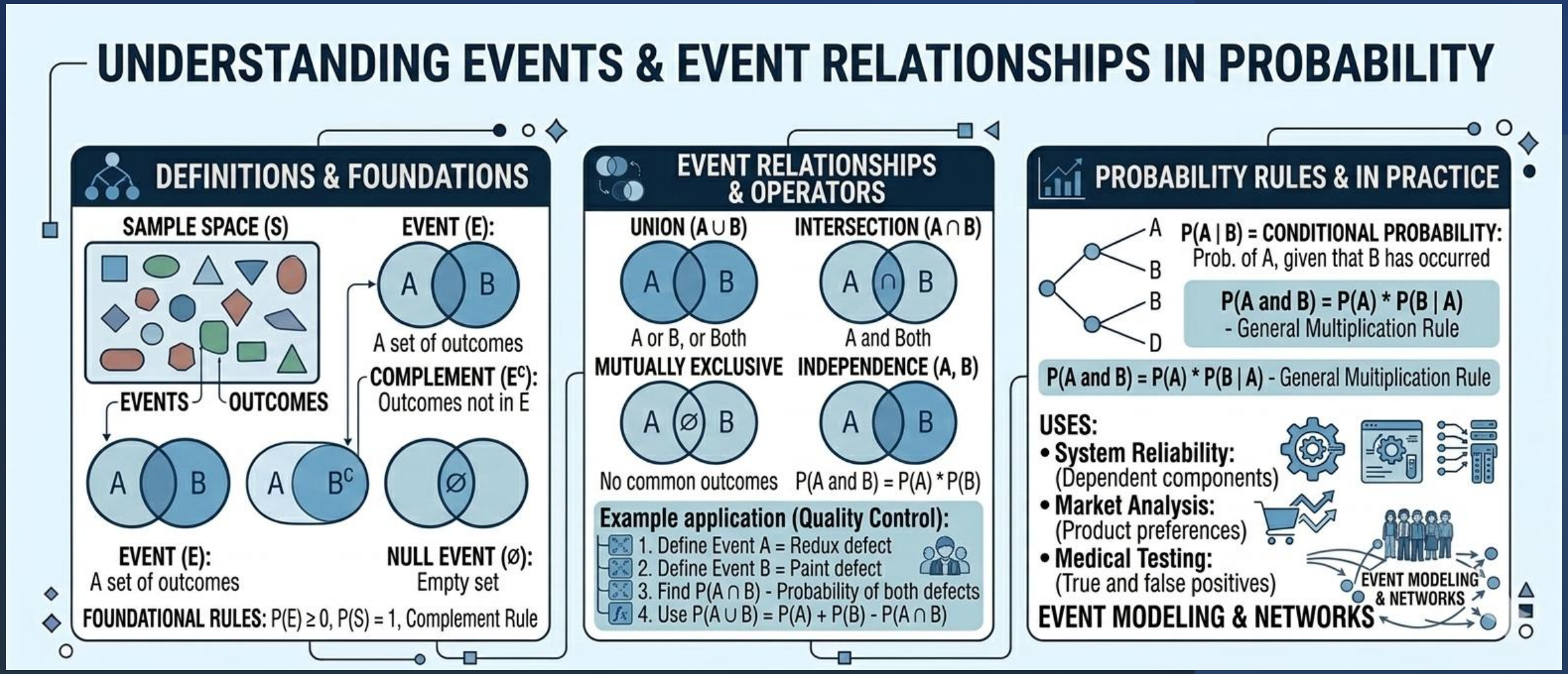


USES:

- **Medical Diagnosis:** (Update disease probability)
- **Spam Filtering:** (Learn word patterns)
- **Fraud Detection:** (Assess abnormal transaction probability)



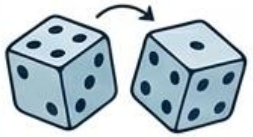
PROBABILITY MODELING & INFERENCE


UNDERSTANDING EVENTS & EVENT RELATIONSHIPS IN PROBABILITY
DEFINITIONS & FOUNDATIONS
SAMPLE SPACE (S)
EVENTS
OUTCOMES
EVENT (E):
A set of outcomes
COMPLEMENT (E^C):
Outcomes not in E
EVENT (E):
A set of outcomes
NULL EVENT (∅):
Empty set
FOUNDATIONAL RULES: P(E) ≥ 0, P(S) = 1, Complement Rule
EVENT RELATIONSHIPS & OPERATORS
UNION (A ∪ B)
A or B, or Both
INTERSECTION (A ∩ B)
A and Both
MUTUALLY EXCLUSIVE
No common outcomes
INDEPENDENCE (A, B)
P(A and B) = P(A) * P(B)
Example application (Quality Control):
1. Define Event A = Redux defect
2. Define Event B = Paint defect
3. Find P(A ∩ B) - Probability of both defects
4. Use P(A ∪ B) = P(A) + P(B) - P(A ∩ B)
PROBABILITY RULES & IN PRACTICE
P(A | B) = CONDITIONAL PROBABILITY:
Prob. of A, given that B has occurred
P(A and B) = P(A) * P(B | A)
- General Multiplication Rule
P(A and B) = P(A) * P(B | A) - General Multiplication Rule
USES:
• System Reliability:
(Dependent components)
• Market Analysis:
(Product preferences)
• Medical Testing:
(True and false positives)
EVENT MODELING & NETWORKS
EVENT MODELING & NETWORKS

INDEPENDENT VS DEPENDENT EVENTS IN STATISTICAL ANALYSIS

DEFINITIONS & FOUNDATIONS

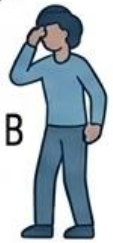
INDEPENDENT EVENTS (A, B):



The outcome of A does not affect the outcome of B

DEPENDENT EVENTS (A | B):

The outcome of A affects the outcome of B



FOUNDATIONAL RULES: Probability P(E) can be defined by frequency over a sample space (S)

Independent	Dependent
(N | A and RI)	-
A	D
B	A

KEY RELATIONSHIPS & TESTS

MULTIPLICATION RULE

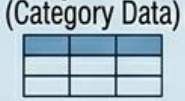
For Independent Events (A and B)	For Dependent Events (A | B)
P(A and B) = P(A) * P(B)	**P(A and B) = P(A) * P(B | A)**
Rule text oples of A doesn't affect the outcome of B	Rule text rules affect A not the outcome of B

CONDITIONAL PROBABILITY

P(A | B) for Independent	P(A | B) for Dependent
P(A | B) = P(A)	**P(A | B) ≠ P(A)**
Tule text or priend	Tule for event

STATISTICAL TESTS

Independent Events	Dependent Events
Chi-Squared Test for Independence (Category Data)	**Correlation Analysis** (Numerical Data)



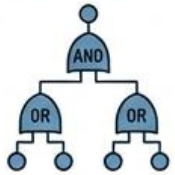
STATISTICAL APPLICATIONS & INFERENCE

GENERAL MULTIPLICATION RULE	CONDITIONAL PROBABILITY
P(A and B) = P(A) * P(B) - General Multiplication Rule - General Multiplication Rule	**P(A and B) = P(A) * P(B | A)** - General Eutcmons General Multiplication Rule

USES:

Medical Studies: (Drug effect vs. placebo)

A/B Testing: (Webpage performance)



System Fault Trees: (Dependent component failures)

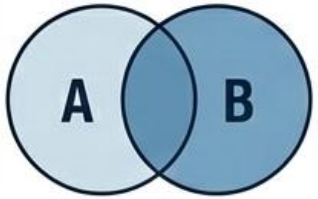
STATISTICAL MODELING & INFERENCE

STATISTICAL MODELING & INFERENCE

CONCEPT OF MUTUALLY EXCLUSIVE EVENTS & THEIR APPLICATIONS

DEFINITIONS & FUNDATIONS

MUTUALLY EXCLUSIVE EVENTS (A, B):



Events that cannot occur simultaneously

Heads/ Tails

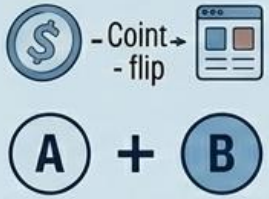
Heads Day/Night

FOUNDATIONAL RULES: Probability P(E) over a sample space (S) for Mutually Exclusive

Mutually Exclusive	Not Mutually Exclusive
P(A and B) = 0	-
P(A or B) = 0	P(A or B) = P(A) + P(B)

KEY RELATIONSHIPS & TESTS

UNION (OR RULE)



P(A or B) = P(A) + P(B)

Rule text of A doesen't affect of doesn't net outcome of B

NULL INTERSECTION

P(A and B) = 0

Rule text of Eudoan't affect of A B not the outcome of B

STATISTICAL TESTS

- **Chi-Squared Test** (for frequency and non-overlapping data)
- **ANOVA** (if events define non-overlapping groups)

APPLICATIONS & INFERENCE

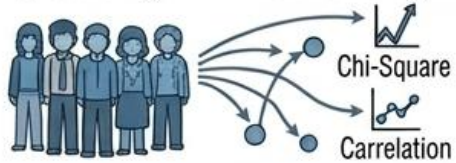
GENERAL MULTIPLICATION RULE	CONDITIONAL PROBABILITY
P(A or B) = P(A) + P(B) - General Multiplication Rule - General Multiplication Rule	**P(A and B) = 0** - General Eutemons General Multiplication Rule

USES:

Coin Flipping: (Heads vs. Tails probability)

Categorical Selection: (Gender, Eye Color, Political Party)

Segmentation: (Selecting single segments)

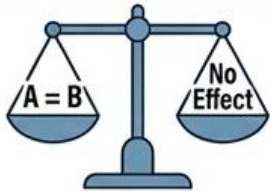


CATEGORICAL MOFERENCE

CATEGORICAL MODELING & INFERENCE

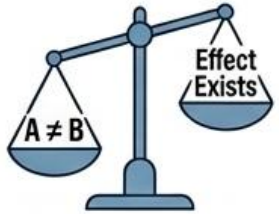
INTRODUCTION TO THE HYPOTHESIS TESTING FRAMEWORK

DEFINITIONS & FOUNDATIONS



NULL HYPOTHESIS (H0): Status quo or standard assumption

THE ALTERNATIVE HYPOTHESIS (H1): A new effect or difference to be proven



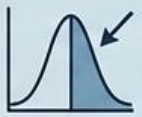
FOUNDATIONAL RULES: Probability P(E) can be defined by frequency, but P(Value > Observed | H_0 is True) is the P-Value

Decision Outcomes	Decision Outcomes
Correct Reject H_0 Correct Fail to Reject H_0	Type I Error (α)
Type I Error (α)	Type II Error (β)

KEY TESTS & RELATIONSHIPS

ONE-TAILED TEST	TWO-TAILED TEST
Tests for directional change	Tests for any difference (directional or non-directional)

STATISTICAL SIGNIFICANCE & TESTS



P-Value and Alpha (α)

- **T-Test** (for continuous data)
- **Chi-Squared Test** (for categorical data)
- **ANOVA** (for multiple group comparison)

APPLICATIONS & INFERENCE

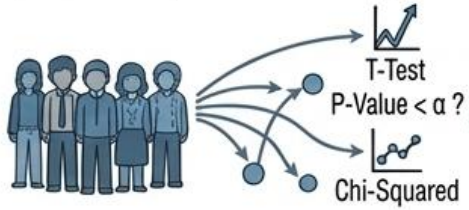
P(Value > Observed | H0) is P-Value,
P(H0 True | Reject H0) is Type I Error,
P(H0 True | Reject H0) is Type I Error..

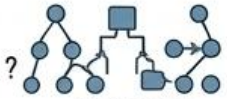
USES:

A/B Testing: (Compare webpage performance)

Quality Control: (Detect production defects)

Medical Trials: (Assess drug effectiveness)



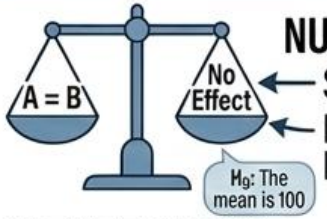


DECISION INFERENCE

DECISION MAKING & STATISTICAL INFERENCE

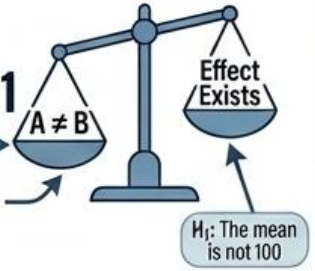
FORMULATING NULL HYPOTHESIS (H_0) & ALTERNATIVE HYPOTHESIS (H_1)

DEFINITIONS & FOUNDATIONS



NULL HYPOTHESIS (H_0):
- Standard Condition
- NO SIGNIFICANT DIFFERENCE OR EFFECT

THE ALTERNATIVE HYPOTHESIS (H_1): H_1



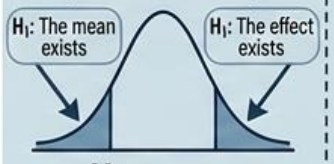
- STATEMENT TO BE PROVEN
- PRESENCE OF DIFFERENCE OR EFFECT

MUTUALLY EXCLUSIVE PARTNERS

STATEMENT TYPE	H_0	H_1
Directional	$H_0: \mu = \mu_0$ $H_0: \mu \le \mu_0$	$H_1: \mu \ne \mu_0$ $H_1: \mu > \mu_0$
Non-directional	$H_0: \mu \ge \mu_0$	$H_1: \mu < \mu_0$

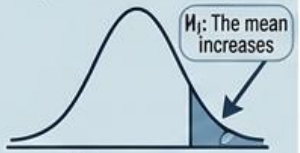
TYPES OF ALTERNATIVE HYPOTHESES & EXAMPLES

NON-DIRECTIONAL (TWO-TAILED)



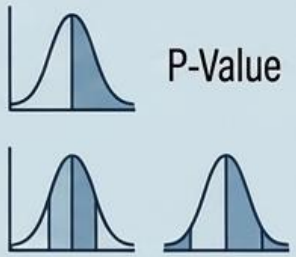
$H_1: \mu \ne \mu_0$
(A is different than B)

DIRECTIONAL (ONE-TAILED)



$H_1: \mu > \mu_0$
(A is greater than B)

FORMULATION & TESTS



P-Value

- P-Value
- T-Test
- Chi-Squared Test
- ANOVA

APPLICATIONS & INFERENCE

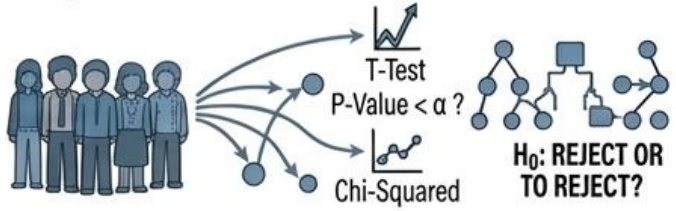
P(Value > Observed | H0) is P-Value,
P(H0 True | Reject H0) is Type I Error,
P(H0 True | Reject H0) is Type I Error..

USES:

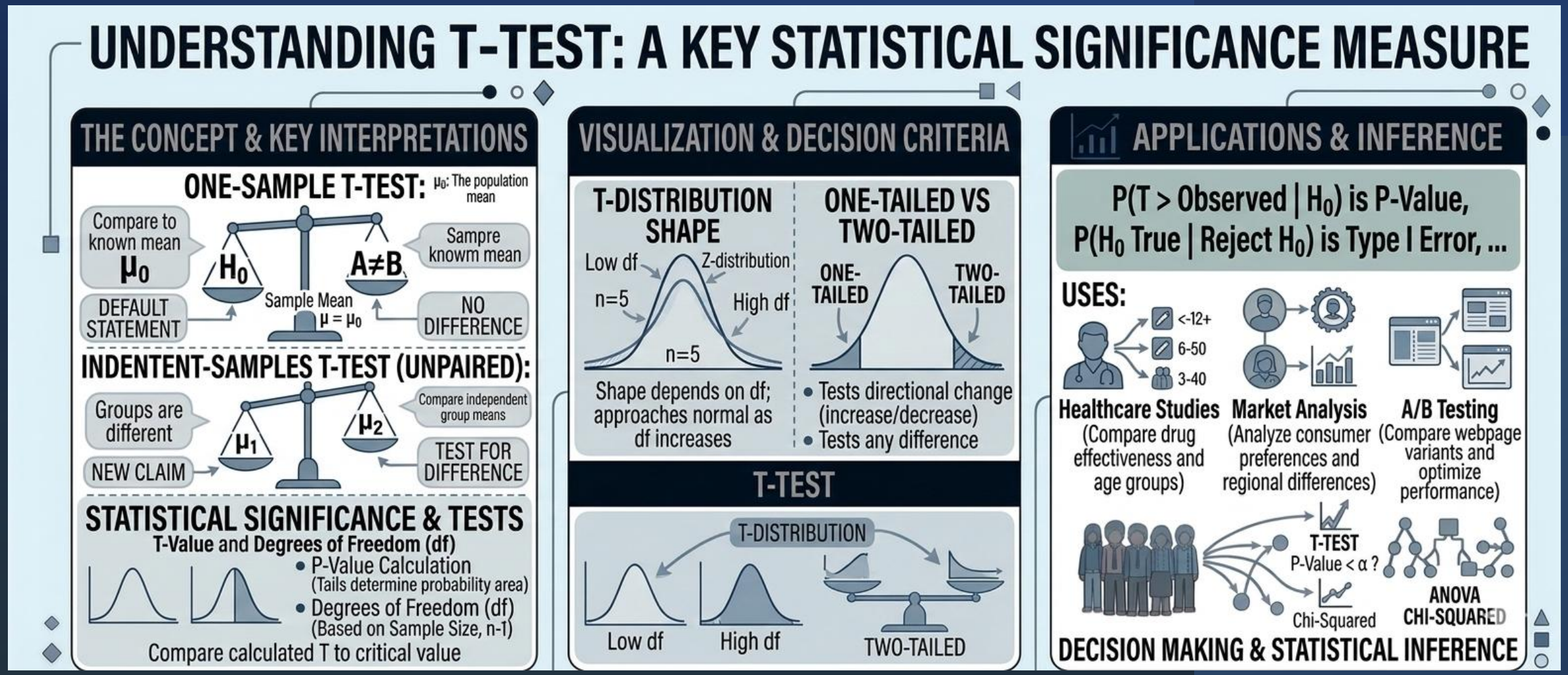
Medical Trials: (Assess a new drug effect)

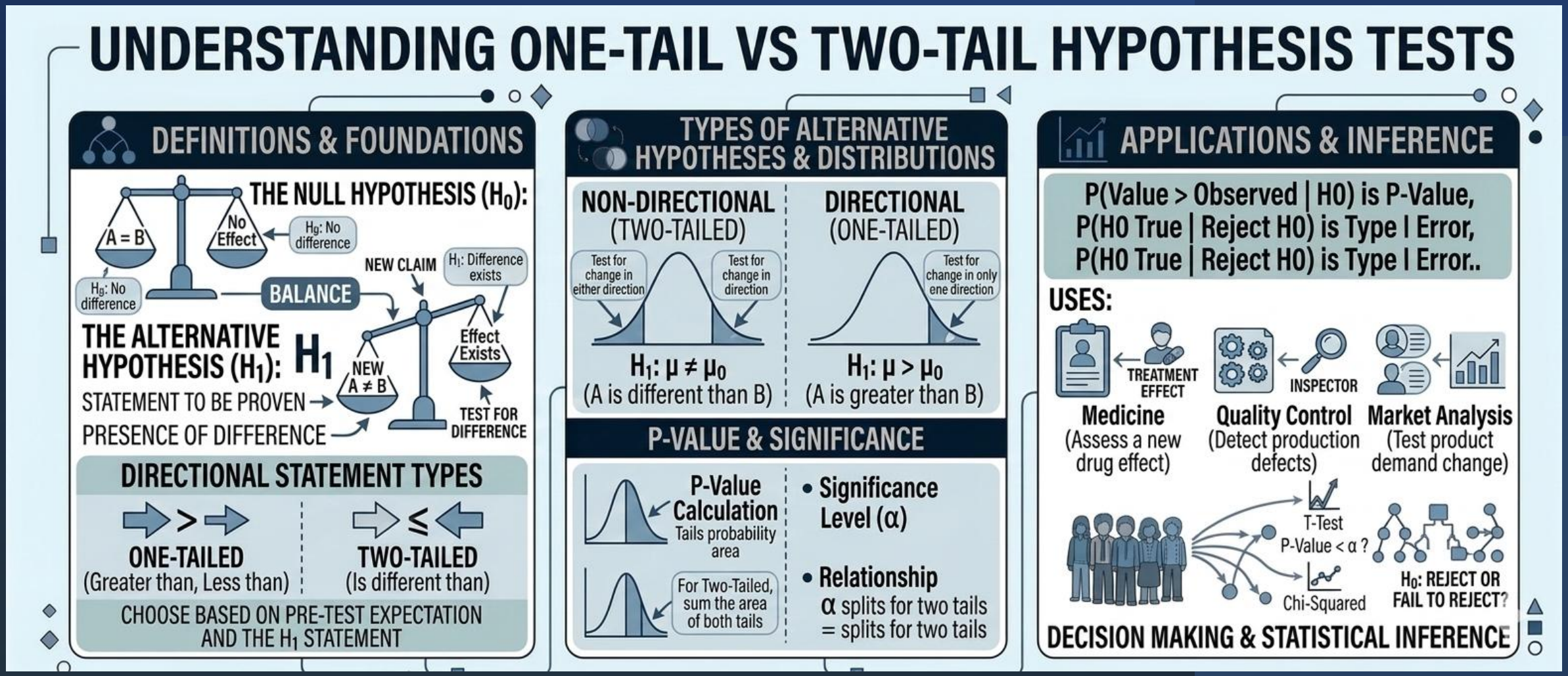
A/B Testing: (Compare webpage variants)

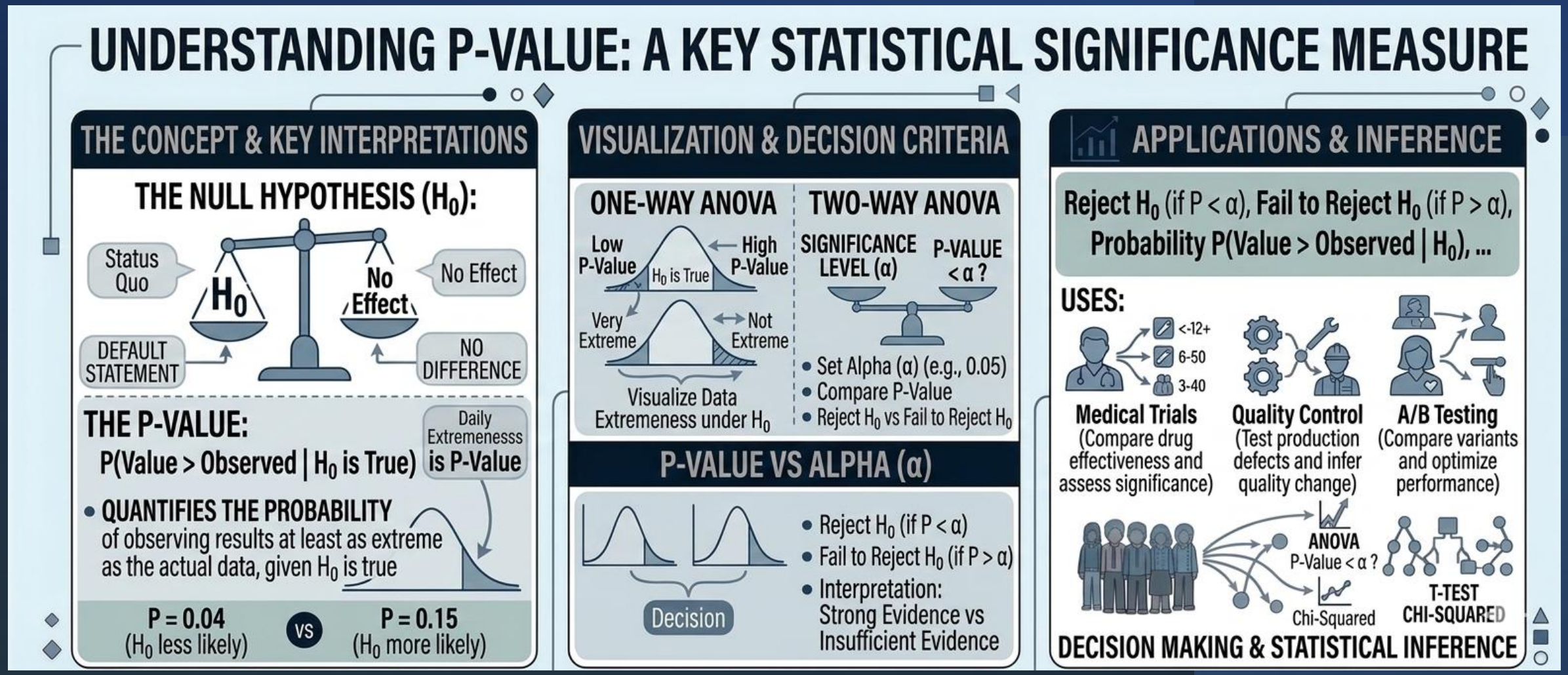
Market Analysis: (Test product demand)

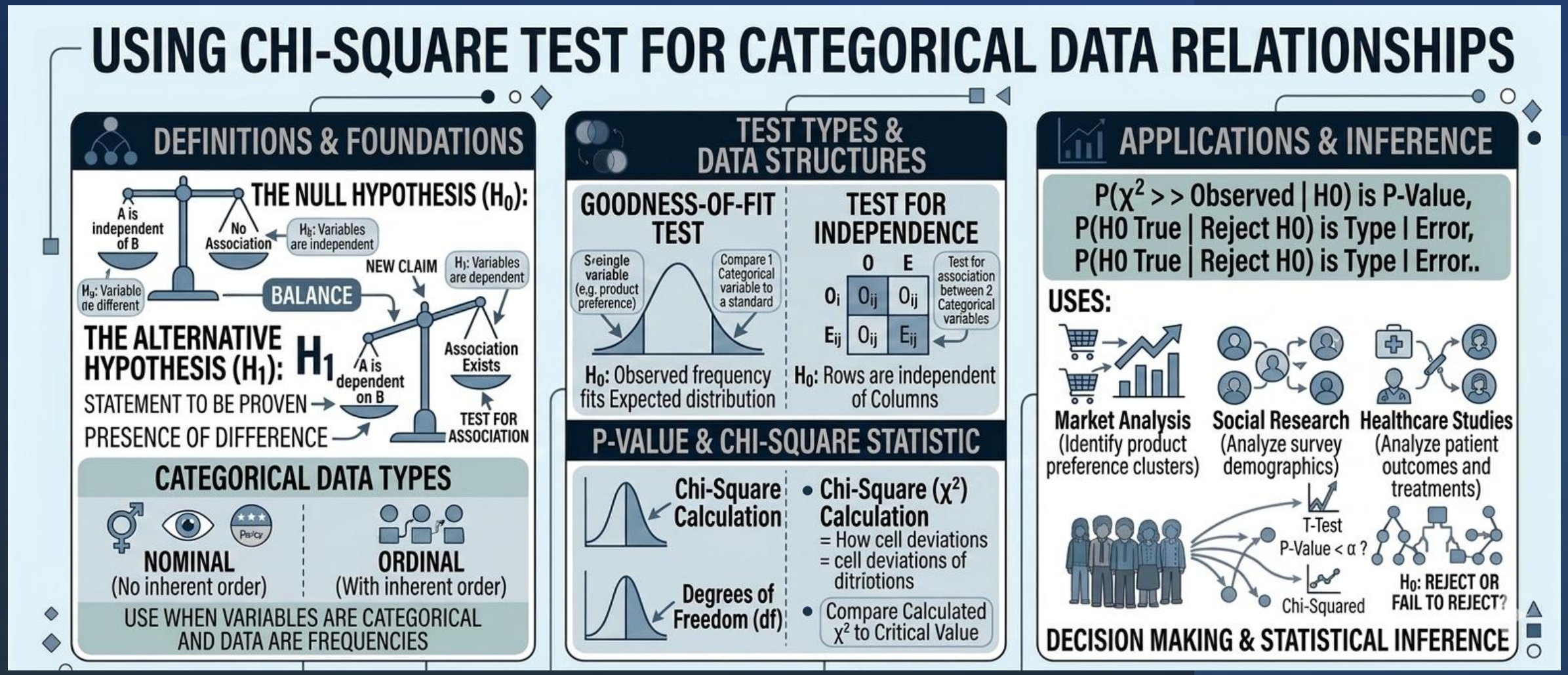


DECISION MAKING & STATISTICAL INFERENCE


UNDERSTANDING T-TEST: A KEY STATISTICAL SIGNIFICANCE MEASURE
THE CONCEPT & KEY INTERPRETATIONS
ONE-SAMPLE T-TEST:
μ0: The population mean
Compare to known mean μ0
H0
A≠B
Sampre knowm mean
DEFAULT STATEMENT
Sample Mean μ = μ0
NO DIFFERENCE
INDENTENT-SAMPLES T-TEST (UNPAIRED):
Groups are different
μ1
μ2
Compare independent group means
NEW CLAIM
TEST FOR DIFFERENCE
STATISTICAL SIGNIFICANCE & TESTS
T-Value and Degrees of Freedom (df)
• P-Value Calculation
(Tails determine probability area)
• Degrees of Freedom (df)
(Based on Sample Size, n-1)
Compare calculated T to critical value
VISUALIZATION & DECISION CRITERIA
T-DISTRIBUTION SHAPE
Low df
Z-distribution
n=5
High df
n=5
Shape depends on df; approaches normal as df increases
ONE-TAILED VS TWO-TAILED
ONE-TAILED
TWO-TAILED
• Tests directional change (increase/decrease)
• Tests any difference
T-TEST
T-DISTRIBUTION
Low df
High df
TWO-TAILED
APPLICATIONS & INFERENCE
P(T > Observed | H0) is P-Value,
P(H0 True | Reject H0) is Type I Error, ...
USES:
<-12+
6-50
3-40
Healthcare Studies
(Compare drug effectiveness and age groups)
Market Analysis
(Analyze consumer preferences and regional differences)
A/B Testing
(Compare webpage variants and optimize performance)
T-TEST
P-Value < α ?
Chi-Squared
ANOVA
CHI-SQUARED
DECISION MAKING & STATISTICAL INFERENCE


UNDERSTANDING ONE-TAIL VS TWO-TAIL HYPOTHESIS TESTS
DEFINITIONS & FOUNDATIONS
THE NULL HYPOTHESIS (H_0):
A = B
No Effect
H_0: No difference
H_0: No difference
BALANCE
NEW CLAIM
H_1: Difference exists
THE ALTERNATIVE HYPOTHESIS (H_1): H_1
NEW A ≠ B
Effect Exists
STATEMENT TO BE PROVEN
PRESENCE OF DIFFERENCE
TEST FOR DIFFERENCE
DIRECTIONAL STATEMENT TYPES
>
≤
ONE-TAILED
(Greater than, Less than)
TWO-TAILED
(Is different than)
CHOOSE BASED ON PRE-TEST EXPECTATION AND THE H_1 STATEMENT
TYPES OF ALTERNATIVE HYPOTHESES & DISTRIBUTIONS
NON-DIRECTIONAL
(TWO-TAILED)
Test for change in either direction
Test for change in direction
H_1: $\mu \neq \mu_0$
(A is different than B)
DIRECTIONAL
(ONE-TAILED)
Test for change in only ene direction
H_1: $\mu > \mu_0$
(A is greater than B)
P-VALUE & SIGNIFICANCE
P-Value Calculation
Tails probability area
For Two-Tailed, sum the area of both tails
• Significance Level (α)
• Relationship
α splits for two tails
= splits for two tails
APPLICATIONS & INFERENCE
P(Value > Observed | H0) is P-Value,
P(H0 True | Reject H0) is Type I Error,
P(H0 True | Reject H0) is Type I Error..
USES:
TREATMENT EFFECT
INSPECTOR
Medicine
(Assess a new drug effect)
Quality Control
(Detect production defects)
Market Analysis
(Test product demand change)
T-Test
P-Value < α ?
Chi-Squared
H_0: REJECT OR FAIL TO REJECT?
DECISION MAKING & STATISTICAL INFERENCE


UNDERSTANDING P-VALUE: A KEY STATISTICAL SIGNIFICANCE MEASURE
THE CONCEPT & KEY INTERPRETATIONS
THE NULL HYPOTHESIS (H_0):
Status Quo
H_0
No Effect
No Effect
DEFAULT STATEMENT
NO DIFFERENCE
THE P-VALUE:
P(Value > Observed | H_0 is True)
Daily Extremenesss is P-Value
• QUANTIFIES THE PROBABILITY of observing results at least as extreme as the actual data, given H_0 is true
P = 0.04 (H_0 less likely)
VS
P = 0.15 (H_0 more likely)
VISUALIZATION & DECISION CRITERIA
ONE-WAY ANOVA
Low P-Value
High P-Value
H_0 is True
Very Extreme
Not Extreme
Visualize Data Extremeness under H_0
TWO-WAY ANOVA
SIGNIFICANCE LEVEL (α)
P-VALUE < α ?
• Set Alpha (α) (e.g., 0.05)
• Compare P-Value
• Reject H_0 vs Fail to Reject H_0
P-VALUE VS ALPHA (α)
Decision
• Reject H_0 (if P < α)
• Fail to Reject H_0 (if P > α)
• Interpretation: Strong Evidence vs Insufficient Evidence
APPLICATIONS & INFERENCE
Reject H_0 (if P < α), Fail to Reject H_0 (if P > α), Probability P(Value > Observed | H_0), ...
USES:
<-12+
6-50
3-40
Medical Trials (Compare drug effectiveness and assess significance)
Quality Control (Test production defects and infer quality change)
A/B Testing (Compare variants and optimize performance)
ANOVA
P-Value < α ?
Chi-Squared
T-TEST
CHI-SQUARED
DECISION MAKING & STATISTICAL INFERENCE


USING CHI-SQUARE TEST FOR CATEGORICAL DATA RELATIONSHIPS
DEFINITIONS & FOUNDATIONS
THE NULL HYPOTHESIS (H_0):
A is independent of B
No Association
H_0: Variables are independent
H_0: Variable are different
BALANCE
NEW CLAIM
H_1: Variables are dependent
THE ALTERNATIVE HYPOTHESIS (H_1): H_1
A is dependent on B
Association Exists
STATEMENT TO BE PROVEN
PRESENCE OF DIFFERENCE
TEST FOR ASSOCIATION
CATEGORICAL DATA TYPES
NOMINAL
(No inherent order)
ORDINAL
(With inherent order)
USE WHEN VARIABLES ARE CATEGORICAL AND DATA ARE FREQUENCIES
TEST TYPES & DATA STRUCTURES
GOODNESS-OF-FIT TEST
Sreingle variable (e.g. product preference)
Compare 1 Categorical variable to a standard
H_0: Observed frequency fits Expected distribution
TEST FOR INDEPENDENCE
O E
O_i O_{ij} O_{ij}
E_{ij} O_{ij} E_{ij}
Test for association between 2 Categorical variables
H_0: Rows are independent of Columns
P-VALUE & CHI-SQUARE STATISTIC
Chi-Square Calculation
Degrees of Freedom (df)
• Chi-Square (χ^2) Calculation
= How cell deviations
= cell deviations of ditriotions
• Compare Calculated χ^2 to Critical Value
APPLICATIONS & INFERENCE
$P(\chi^2 >> \text{Observed} \mid H0)$ is P-Value,
$P(H0 \text{ True} \mid \text{Reject } H0)$ is Type I Error,
$P(H0 \text{ True} \mid \text{Reject } H0)$ is Type I Error..
USES:
Market Analysis
(Identify product preference clusters)
Social Research
(Analyze survey demographics)
Healthcare Studies
(Analyze patient outcomes and treatments)
T-Test
P-Value < α ?
Chi-Squared
H_0: REJECT OR FAIL TO REJECT?
DECISION MAKING & STATISTICAL INFERENCE

APPLYING ANOVA (ANALYSIS OF VARIANCE) FOR MULTI-GROUP COMPARISON

DEFINITIONS & FOUNDATIONS

THE NULL HYPOTHESIS (H_0):

H_0: All group means are equal ($\mu_1=\mu_2=...=\mu_k$)

A single condition

H_0: N condition (standard)

H_0: All group means equal

BALANCE

NEW CLAIM

H_1: Difference exists

THE ALTERNATIVE HYPOTHESIS (H_1): H_1

H_1: At least one group mean is different

TEST FOR DIFFERENCE

STATEMENT TO BE PROVEN

CATEGORICAL INDEPENDENT VARIABLES

Example	Time Point	Age Group
Treatment Group	[..., ...], ...]	($\mu_1 - \mu_k$)
Time Point	[..., ...], ...]	($\mu_2 - \mu_k$)
Age Group	[..., ...], ...]	...
$K \geq 3$ groups	[..., ...]	...

USE FOR NUMERICAL DATA OVER MUTUALLY EXCLUSIVE GROUPS

TEST TYPES & DATA STRUCTURES

ONE-WAY ANOVA

μ_1 μ_2 μ_3

Analyze 1 Independent Variable over 3+ Groups

H_0: Group means fit Expected distribution

TWO-WAY ANOVA

Column effect

Row effect | Column effect | Interaction

	x_1	x_2	x_3
	x_1	x_2	x_3
	x_1	x_2	x_3

Test for 2 Independent Variables (and their interaction)

H_0: Groups have independent Row/Col effects

P-VALUE & F-STATISTIC

F-Statistic Calculation

Degrees of Freedom (df)

- **Between-Group** – Variance components **Within-Group**
- Exp: $\mu_1, \mu_2, \mu_3, \mu_3$ == ($x_1, x_1, x_2, x_3, ...$)
- Compare Calculated F to Critical Value

APPLICATIONS & INFERENCE

P(F > Observed | H_0) is P-Value,
P(H0 True | Reject H_0) is Type I Error,
P(H0 True | Reject H_0) is Type I Error.

USES:

<-12+
6-50
3-40

Medical Trials (Compare 3+ drug effectiveness and age)

Quality Control (Analyze production line variations)

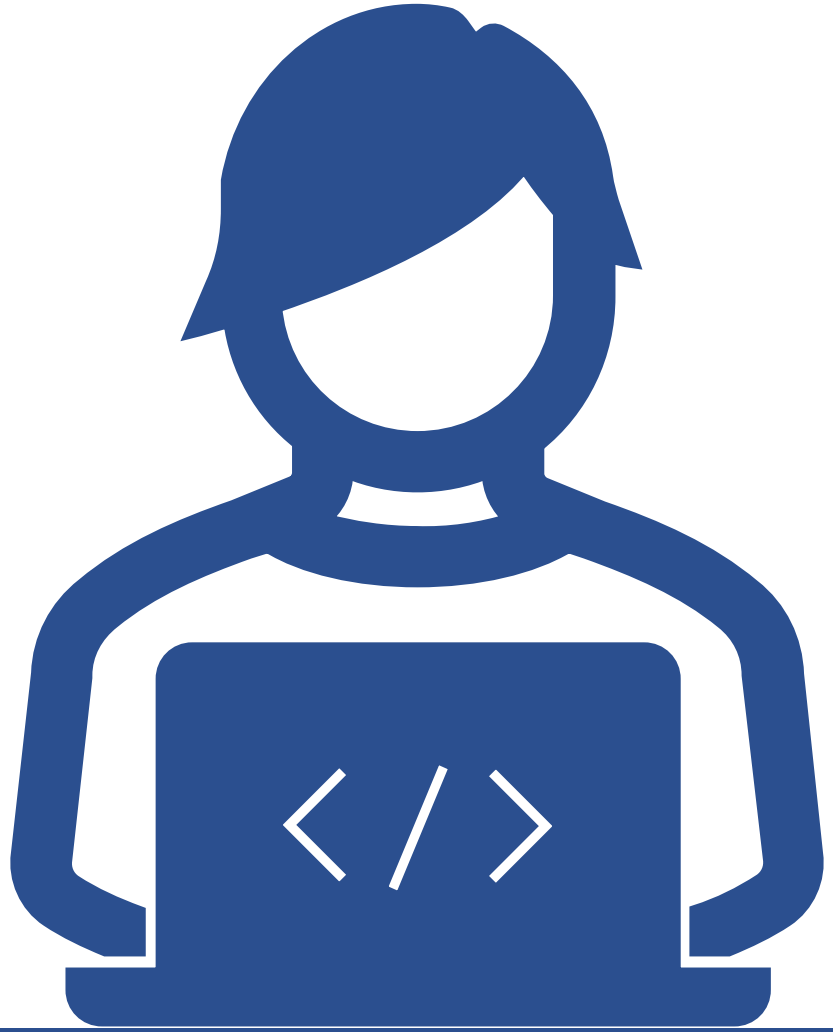
Market Research (Test consumer preferences across segments)

T-Test
P-Value < α ?
Chi-Squared

H_0: REJECT OR FAIL TO REJECT?

DECISION MAKING & STATISTICAL INFERENCE

Hands - on



Summary

- Participants understood how **probability concepts help model uncertainty and support data-driven decision making**.
- The session introduced **different types of event relationships** including independent, dependent, and mutually exclusive events.
- Learners explored the **hypothesis testing framework**, including null vs alternative hypotheses and one-tail vs two-tail testing approaches.
- Statistical techniques such as **Chi-Square and ANOVA** were discussed for analyzing categorical relationships and comparing multiple groups in real-world datasets.

Any Query

Thank You

Happy learning